SMART INDIA HACKATHON 2026
PROJECT REPORT

HeatSafe AI

AI-Powered Heatwave Early Warning & Human Thermal Stress Monitoring Platform

Problem Statement ID	SIH26083
Problem Statement Title	Extreme Heatwave Early Warning and Human Thermal Stress Index
Theme	Disaster Management
Category	Software
Sponsoring Ministry / Organisation	Ministry of Earth Sciences (MoES)
Team Name	[Enter Team Name]
Team Leader	[Enter Team Leader Name]
Repository	github.com/GurmeetSingh124/SIH-26083

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1. Introduction

1.1 Background

India is among the countries most severely affected by extreme heat. Heatwaves in the Indo-Gangetic plains and northern states routinely push daytime temperatures past 45°C, and rising humidity in many regions now compounds the danger through elevated "feels-like" temperatures. Outdoor workers, farmers, the elderly, and children are the most vulnerable groups, and existing heat-alert systems are often generic, city-level advisories that do not reflect hyper-local conditions or an individual's real-time exposure.

Problem Statement SIH26083, issued by the Ministry of Earth Sciences (MoES) under the Disaster Management theme, calls for a software solution capable of Extreme Heatwave Early Warning and a Human Thermal Stress Index — a system that goes beyond raw temperature readings to model how the human body actually experiences heat, and that can warn people before conditions become dangerous.

1.2 Problem Statement Summary

- Build an early-warning system for extreme heatwave events using live meteorological data.
- Compute a Human Thermal Stress Index that reflects real physiological heat risk, not just air temperature.
- Translate the index into clear, actionable guidance for citizens, outdoor workers, and disaster-management authorities.
- Make the system usable at the hyper-local (location-specific) level rather than broad city-wide averages.

1.3 Our Response

HeatSafe AI is a full-stack web platform that ingests live weather data for any location, runs it through a trained machine-learning risk-classification model together with a validated heat-index formula, and returns an easy-to-understand four-tier risk level with a concrete recommended action — refreshed continuously as conditions change.

2. Proposed Solution

HeatSafe AI combines live weather telemetry, a supervised machine-learning risk model, geospatial visualisation, and a conversational safety assistant into one dashboard. The platform is built as a decoupled React (Vite) single-page application talking to a FastAPI backend over a REST API, with MongoDB for user accounts and saved locations.

2.1 Core Idea

Rather than displaying only a temperature reading, HeatSafe AI fuses four live meteorological variables — temperature, humidity, wind speed, and solar radiation — into a single, explainable Heat Risk Level (Green / Yellow / Orange / Red). Each level is paired with a risk score, a heatwave probability, a model-confidence figure, plain-language reasons, and a specific recommended action, so the output is meaningful to a non-technical citizen as well as to a disaster-management operator.

2.2 Who It Serves

- Citizens and outdoor workers who need a simple, location-aware heat-safety reading before stepping outdoors.
- Farmers and daily-wage labourers whose work hours can be planned around forecast risk windows.
- District disaster-management authorities who need a live heat-zone map to prioritise interventions (water points, ORS camps, health-centre readiness).
- Elderly residents and caregivers who need advance notice ahead of extreme (Red) conditions.

3. System Architecture

The system follows a three-layer architecture — client, application, and data/external — which keeps the ML risk engine, authentication, and third-party integrations cleanly separated behind a single FastAPI gateway.

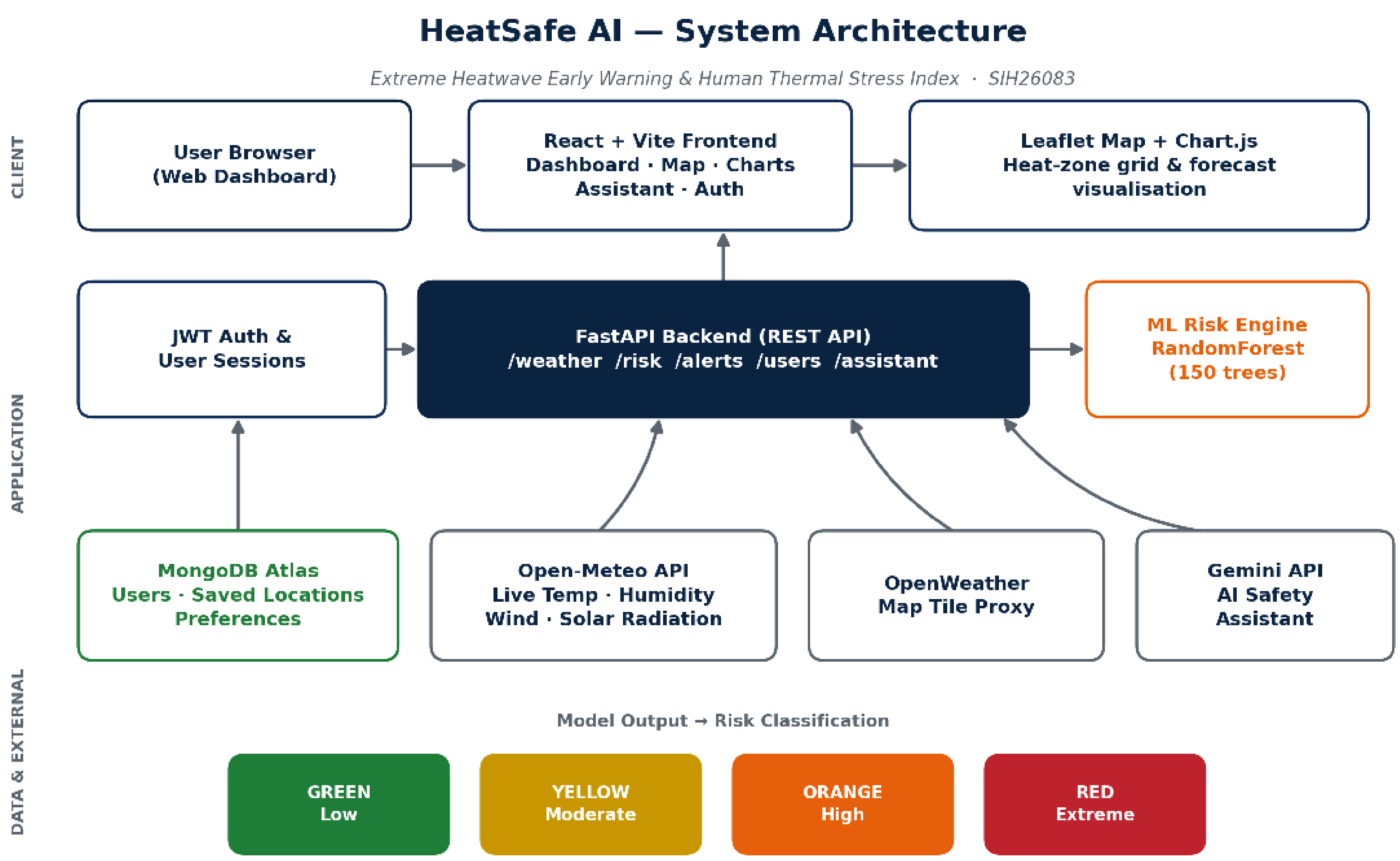

Figure 1: HeatSafe AI system architecture — client, application, and data/external layers.

3.1 Layer Description

- Client layer: A React + Vite dashboard renders live weather cards, the risk gauge, a Leaflet map of heat zones, Chart.js forecast graphs, and the AI assistant chat panel.
- Application layer: A single FastAPI service exposes REST endpoints for weather, risk prediction, alerts, user accounts, and the AI assistant, and hosts the ML risk engine in-process for low-latency predictions.
- Data & external layer: MongoDB Atlas stores user accounts and saved locations; Open-Meteo supplies free live weather and forecast data; OpenWeatherMap tiles are proxied through the backend to keep the API key server-side; Gemini powers the optional conversational safety assistant.

3.2 Design Principles

- No API keys in the browser — all third-party calls (weather tiles, Gemini) are proxied through the backend.

- Zero-cost core data path — Open-Meteo requires no API key, keeping the live weather and forecast pipeline free to run at scale.
- Stateless prediction — the ML model is a pure function of four weather variables, making it fast, cache-friendly, and easy to redeploy.

4. Technology Stack

Layer	Technology	Purpose
Frontend	React 18 + Vite 5	Component-based SPA dashboard
Mapping	Leaflet + React-Leaflet	Heat-zone grid map visualisation
Charts	Chart.js + react-chartjs-2	Forecast & trend visualisation
HTTP client	Axios	Frontend ↔ backend communication
Backend API	FastAPI (Python)	Async REST API layer
ML runtime	scikit-learn 1.7.2, joblib, numpy, pandas	Risk model training & inference
Database	MongoDB (Atlas or local) + Motor/PyMongo	Users, saved locations, preferences
Auth	JWT (python-jose) + bcrypt	Secure user sessions
Weather data	Open-Meteo API (free, no key)	Live weather + hourly forecast
Map tiles	OpenWeatherMap tile API (proxied)	Temperature / cloud / wind overlays
AI assistant	Google Gemini API	Conversational heat-safety guidance
Build/Deploy	npm / Vite build, Uvicorn ASGI server	Production frontend build & API serving

5. Machine Learning Model

At the heart of HeatSafe AI is a RandomForestClassifier (150 estimators, max depth 8, scikit-learn 1.7.2) trained to classify live weather conditions into one of four heat-risk levels. Its prediction is cross-validated in real time against the Rothfusz heat-index regression — the same formula used by meteorological agencies to compute the “feels-like” temperature — so the final output is both data-driven and physically grounded.

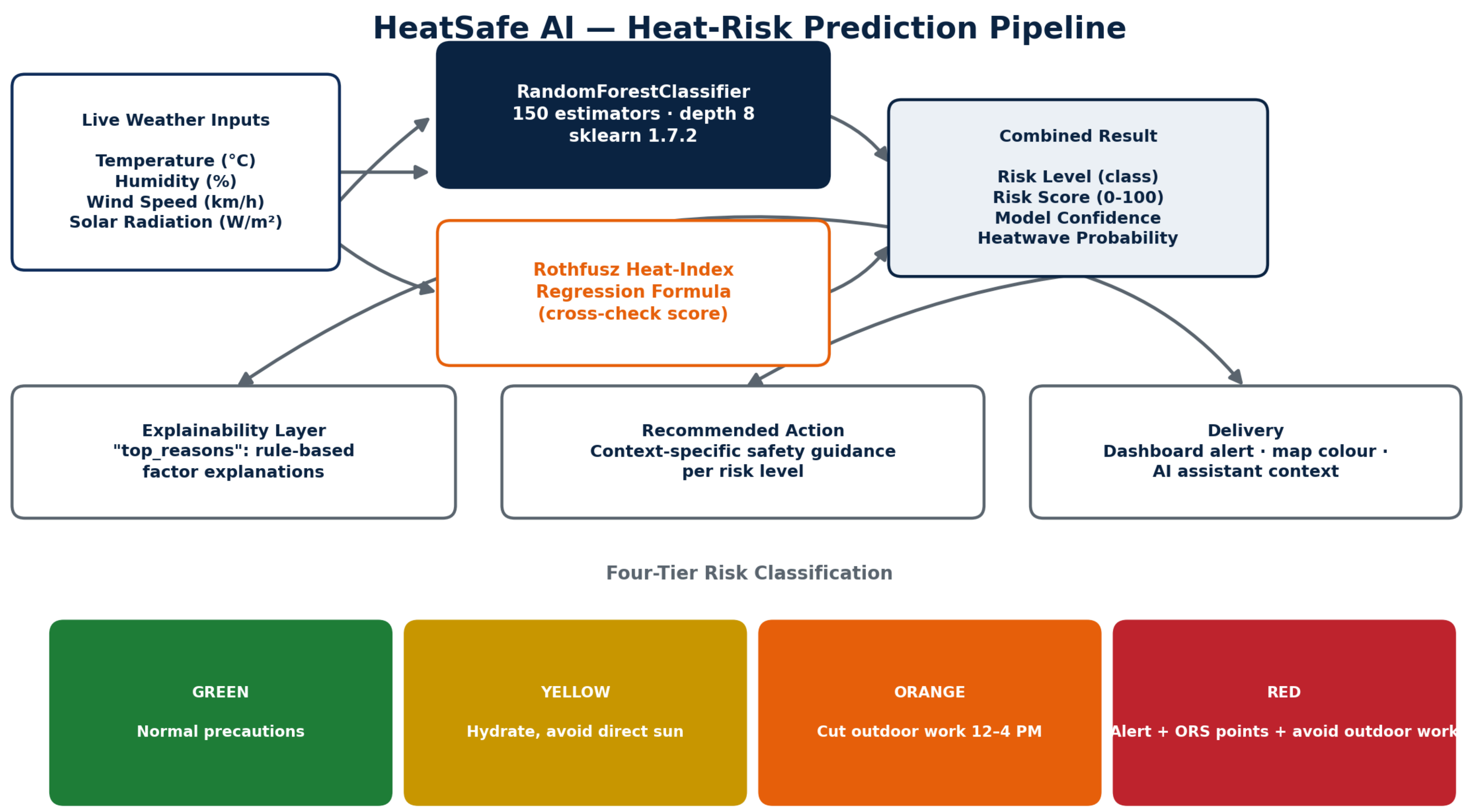

Figure 2: End-to-end heat-risk prediction pipeline, from live weather inputs to the delivered alert.

5.1 Input Features

Feature	Unit	Description
temperature_c	˚C	Live ambient air temperature
humidity_pct	%	Relative humidity
wind_speed_kmh	km/h	Wind speed (low wind intensifies heat stress)
solar_radiation_w_m2	W/m^2	Solar radiation load

5.2 Output

Risk Level	Status	Example Trigger	Recommended Action
Green	Low	Normal range conditions	Routine precautions, stay hydrated
Yellow	Moderate	Rising temperature / humidity	Hydrate, limit direct sun exposure
Orange	High	Temp ≥ 35– 40˚C, high humidity	Reduce outdoor work between 12 PM–4 PM
Red	Extreme	Temp ≥ 40˚C + low wind + high solar load	Immediate alert, ORS/water points, avoid outdoor work

5.3 Explainability

Every prediction ships with a set of plain-language “top reasons” (e.g. “temperature is very high”, “low wind speed will make heat feel worse”) and a model-confidence percentage, so end users and disaster-management staff can see why an alert was raised, not just the raw classification.

5.4 Derived Metrics

- Heat Index (°C): computed via the Rothfusz regression as a cross-check against the classifier.
- Risk Score (0–100): a normalised severity score derived from the heat index.
- Heatwave Probability: the combined predicted probability of the Orange + Red classes.
- Model Confidence: the classifier's maximum class probability for the current reading.

6. Key Features

6.1 Monitoring & Prediction

- Live weather ingestion for any latitude/longitude via Open-Meteo, with no API key required.
- Real-time four-tier heat-risk classification (Green / Yellow / Orange / Red) via the trained ML model.
- 6–72 hour heatwave risk forecast chart, so users can plan ahead rather than react.
- Grid-based heat-zone map showing risk levels across a configurable radius around a location.

6.2 Alerts & Guidance

- Automatic alert generation with severity, title, message, and the applicable peak-risk time window.
- Context-specific recommended actions tailored to each risk level.
- Optional Gemini-powered AI assistant that answers heat-safety questions using the user’s live risk and weather context.

6.3 Personalisation & Accounts

- Location search and browser-geolocation support for instant hyper-local readings.
- Secure JWT-based registration/login and saved-locations list (“My Area”) for quick re-checks.
- Weather-map overlays (temperature, clouds, precipitation, wind) proxied securely through the backend.

7. API Endpoints

The backend exposes a documented REST API (interactive Swagger UI at /docs) organised into five resource groups:

Method	Endpoint	Purpose
GET	/api/weather/current	Live weather for a given lat/lon
GET	/api/weather/geocode	Search a place name to get coordinates
GET	/api/weather/tiles/{layer}/{z}/{x}/{y}.png	Proxied OpenWeatherMap map tiles
GET	/api/risk/predict	ML-based heat-risk prediction for a location

Method	Endpoint	Purpose
POST	/api/risk/predict-manual	Prediction from manually supplied weather values
GET	/api/risk/heatzones	Grid of risk predictions around a centre point (map layer)
GET	/api/risk/forecast	6–72 hour forward risk forecast
GET	/api/alerts/current	Ready-to-display alert (title, message, severity)
POST	/api/users/...	Registration, login, saved locations
POST	/api/assistant/chat	Conversational Q&A grounded in live risk/weather context

8. User Workflow

- 1. User opens the dashboard; the app requests browser geolocation or a manually searched location.
- 2. The frontend calls /api/weather/current and /api/risk/predict for that location.
- 3. The backend fetches live weather from Open-Meteo, runs it through the ML risk engine, and returns risk level, score, confidence, and recommended action.
- 4. The dashboard renders the risk card, contributing-factors card, and an alert banner if conditions are Orange or Red.
- 5. The user can view the heat-zone map (/api/risk/heatzones) or the forecast chart (/api/risk/forecast) for forward planning.
- 6. The user may ask the AI assistant a question; the assistant answers using the same live risk/weather context.
- 7. Registered users can save frequently checked locations to "My Area" for one-tap access next time.

9. Feasibility & Viability

9.1 Technical Feasibility

- Built entirely on free, open-source, and no-key data sources (Open-Meteo) for the core prediction path, keeping recurring costs low.
- FastAPI + React is a well-documented, widely supported stack, easing handover to government IT teams and long-term maintenance.
- The RandomForest model is lightweight (single .pkl file) and runs inference in milliseconds, suitable for real-time use even on modest hosting.

9.2 Operational Feasibility

- Default location fallback (e.g. Nabha, Punjab) ensures the system remains usable even without precise geolocation.
- A documented Swagger UI (/docs) simplifies integration testing for partner agencies such as NDMA/State Disaster Management Authorities.

9.3 Risks & Mitigation

Risk	Impact	Mitigation
Third-party weather API downtime	Missing live readings	Fallback provider hooks (OpenWeather/Visual Crossing) already scaffolded in config
Low-connectivity rural areas	Users cannot load dashboard	Planned SMS/IVR alert channel (see Future Scope)
Model drift as climate patterns shift	Reduced prediction accuracy over time	Periodic retraining pipeline using logged live readings

10. Impact & Benefits

- Enables early, location-specific heatwave warnings that can reduce heat-related illness and mortality.
- Gives disaster-management authorities a live, map-based view of emerging high-risk zones to prioritise water points, ORS camps, and cooling shelters.
- Supports outdoor workers and farmers in re-scheduling strenuous activity away from peak-risk windows.
- Plain-language, explainable output makes the tool accessible to non-technical citizens, not just meteorologists.
- Free core data pipeline makes the platform viable for nationwide, low-cost deployment.

11. Future Scope

- SMS/IVR-based alerts for users without smartphone or internet access, extending reach to rural and low-connectivity regions.
- Integration with IMD/NDMA official data feeds for authoritative validation and wider institutional adoption.
- Push notifications and a Progressive Web App (PWA)/offline mode for field workers.
- Wearable-device integration (heart rate, body temperature) for individualised thermal-stress risk.
- Multilingual AI assistant support for regional-language safety guidance.
- Historical analytics dashboard for district authorities to study seasonal heatwave trends.

12. Conclusion

HeatSafe AI directly addresses SIH26083 by converting live meteorological data into an explainable, four-tier Human Thermal Stress Index using a trained machine-learning model cross-checked against a scientifically validated heat-index formula. The platform is built on a free, scalable, open-source stack, exposes a fully documented API, and pairs its predictions with concrete, actionable safety guidance — making it practical to pilot at district level and extend nationwide under the Ministry of Earth Sciences' Disaster Management mandate.

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